

Database model documentation

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1. Model details

Model name:

Copy of AgroConecta Physical Export

Version:

2.4

Database engine:

MySQL

Description:

El modelo lógico propuesto cumple con los requerimientos de AgroConecta al permitir la gestión de proveedores, productos, clientes y pedidos. Asimismo, utiliza tablas asociativas para resolver correctamente las relaciones muchos a muchos entre proveedores-productos y pedidos-productos, garantizando la integridad de los datos

2. Tables

2.1. Table supplier

Description:

contains supplier information

2.1.1. Columns

Column name	Type	Properties	Description
id_supplier	int	PK	Unique identifier of the supplier. Primary Key.
farm_name	varchar(100)		Name of the farm or agricultural field.
farmer_name	varchar(100)		Name of the responsible farmer.
ruc	varchar(11)		Unique tax identification number of the supplier.
phone	varchar(15)		Supplier contact phone number.
region	varchar(50)		Region where the supplier operates.

2.2. Table Product

Description:

Product content

2.2.1. Columns

Column name	Type	Properties	Description
id_product	int	PK	Unique identifier of the product. Primary Key.
product_name	varchar(100)		Commercial name of the product.
description	varchar(255)	null	Brief description of the product.
categoria	varchar(50)		description: Descripción breve.
unit_measure	varchar(20)		Measurement unit used for sale.
id_categoria	int		References the category to which the product belongs. Foreign Key.

2.3. Table Supplier_Product

Description:

An associative table that resolves the many-to-many relationship between suppliers and products. It allows you to record which supplier offers each product and the corresponding price.

2.3.1. Columns

Column name	Type	Properties	Description
id_supplier	int	PK	References the supplier. Foreign Key.
id_product	int	PK	References the product. Foreign Key.
price	decimal(10,2)		Product price offered by the supplier.
product_id_product	int		id_producto: Unique identifier of the product.
proveedor_id_proveedor	int		id_proveedor: Unique identifier of the supplier.

2.4. Table Client

Description:

It stores information about registered users who make purchases on the platform.

2.4.1. Columns

Column name	Type	Properties	Description
id_client	int	PK	Unique identifier of the client. Primary Key.
first_name	varchar(100)		Client's first name.
last_name	varchar(100)		Client's last name.
dni	varchar(8)		National identity document number.
email	varchar(100)		Client's email address.
phone	varchar(15)		Client's mobile phone number.

2.5. Table Address

Description:

Save the delivery addresses associated with each customer. A customer can register one or more addresses.

2.5.1. Columns

Column name	Type	Properties	Description
id_address	int	PK	Unique identifier of the address. Primary Key.
id_client	int		id_cliente: Customer who owns the address.
street	varchar(100)		Street name.
number	varchar(10)		House or building number.
district	varchar(50)		Delivery district.
reference	varchar(255)	null	Additional delivery reference.
client_id_client	int		id_cliente: Unique identifier of the client.

2.6. Table order

Description:

It represents the purchase orders placed by customers within the platform.

2.6.1. Columns

Column name	Type	Properties	Description
id_order	int	PK	Unique identifier of the order. Primary Key.
id_client	int		id_cliente: Customer who made the purchase.
order_date	datetime		Date and time when the order was placed.
status	varchar(30)		Current order status.
client_id_client	int		id_cliente: Unique identifier of the client.

2.7. Table Order_Detail

Description:

It records the products included in each order. It allows you to store the quantity purchased and the product's historical price at the time of the transaction.

2.7.1. Columns

Column name	Type	Properties	Description
id_detail	int	PK	Unique identifier of the order detail. Primary Key.

order_id	int	PK	id_pedido: Associated order.
product_id	int		id_producto: Product purchased.
quantity	decimal(10,2)		Quantity requested by the client.
unit_price	decimal(10,2)		Historical product price at purchase time.
order_id_order	int		order_id: Unique identifier of the order.
supplier_product_supplier_id	int		id_proveedor: Supplier reference.
supplier_product_product_id	int		id_producto: Product reference.

2.8. Table Category

Description:

The category table stores the categories used to classify the products offered on the AgroConecta platform. Its purpose is to organize the product catalog in a structured way and facilitate searches, queries, and reports.

2.8.1. Columns

Column name	Type	Properties	Description
id_category	int	PK	Unique identifier of the product category. Primary Key.
category_name	varchar(50)		Name of the category (Fruit, Vegetable, Tuber, etc.).

3. References

3.1. Reference product_supplier

Description:

The supplier_product table is related to the product table through the id_product field, which acts as a foreign key. This relationship allows you to identify which products are part of the offers registered by the suppliers.

Product	0..*	Supplier_Product
id_product	<->	product_id_product

3.2. Reference customer_address

Description:

The address table is related to the customer table through the customer_id field, which acts as a foreign key. This relationship allows storing one or more delivery addresses associated with a customer registered in AgroConecta.

Client	0..*	Address
id_client	<->	client_id_client

3.3. Reference customer_order

Description:

The order table is related to the customer table through the customer_id field, which acts as a foreign key. This relationship allows the identification of which customer placed each order within the AgroConecta platform.

Client	0..*	order
id_client	<->	client_id_client

3.4. Reference supplier_product_supplier

Description:

The supplier_product table is an associative entity that allows you to manage the relationship between suppliers and the products offered on the AgroConecta platform. Its main function is to record which supplier provides each product and the specific price at which they offer it.

supplier	0..*	Supplier_Product
id_supplier	<->	proveedor_id_proveedor

3.5. Reference order_details_order

Description:

The order_detail table is related to the order table through the order_id field, which acts as a foreign key. This relationship allows the storage of the products that are part of each order placed by a customer.

order	0..*	Order_Detail
id_order	<->	order_id_order

3.6. Reference product_order_details_supplier_product

Description:

The order_detail table is related to the product_supplier table through a composite foreign key consisting of supplier_id and product_id. This relationship allows for the identification of exactly which supplier provided each product included in an order.

Supplier_Product	0..*	Order_Detail
id_supplier	<->	supplier_product_supplier_id
id_product	<->	supplier_product_product_id

3.7. Reference product_category

Description:

The product table is related to the category table through the id_category field, which acts as a foreign key. This relationship allows each product to be classified within a specific category in the AgroConecta catalog.

Category	0..*	Product
id_category	<->	id_categoria

4. Areas

4.1. Area_1 subject area

4.1.1. Tables

- [supplier](#)
- [Product](#)
- [Client](#)
- [Category](#)

4.1.2. References

- [product_supplier](#)
- [customer_address](#)
- [customer_order](#)
- [supplier_product_supplier](#)
- [order_details_order](#)
- [product_order_details_supplier_product](#)
- [product_category](#)

4.2. Area_2 subject area

4.2.1. Tables

- [Supplier_Product](#)
- [Address](#)
- [order](#)

4.2.2. References

- [product_supplier](#)
- [customer_address](#)
- [customer_order](#)
- [supplier_product_supplier](#)
- [order_details_order](#)
- [product_order_details_supplier_product](#)
- [product_category](#)

4.3. Area_3 subject area

4.3.1. Tables

- [Order_Detail](#)

4.3.2. References

- [product_supplier](#)
- [customer_address](#)
- [customer_order](#)

- supplier_product_supplier
- order_details_order
- product_order_details_supplier_product
- product_category